

NAME_____SIGN_____

UGANDA CERTIFICATE OF EDUCATION

CHEMISTRY DEPARTMENT

SENIOR TWO

END OF TERM I 2026

DURATION 1:30MINUTES

**INSTRUCTION: ATTEMPT ALL ITEMS IN SECTION A AND ONE ITEM IN
SECTION B**

SECTION A

ATTEMPT ALL ITEMS IN THIS SECTION

Item 1

Rusting is a chemical process that creates reddish-brown coatings on iron or steel materials through oxidations and it's favored by certain conditions.

Mr. Okello, used uncoated roofing materials made from steel since they were shinning, attractive and cheap. However, after some months, the roofing materials started developing a reddish-brown coating and they all changed the colour.

He is now confused and he's asking himself what could have caused this and how can he minimize it. His friend has told him that this is a chemical process and it requires someone with chemistry knowledge to give him more advise. He has decided to contact you.

Task

As a learner of chemistry,

- (a) State the factors that favored the development of reddish-brown coatings on the roofing materials

- (b) State the properties of one of the factors mentioned in (a) above.

(c) Give the use of one of the factors stated in (a) above

(d) Explain the side effect of the reddish-brown coating of the materials and how it can be solved.

Item 2

Aluminium is one of the important elements of the periodic table used in everyday life including making of aluminium foil which is a packaging material

Okwar, a resident of Bugumba, wishes to start-up a small-scale business to distribute aluminium products to the nearby population. However, he's concerned about the nature, suitabilities, the environmental impact of the element and how it can be overcome. He has now decided to contact you since you have more knowledge about the element.

Task

As a learner of chemistry;

(a) Explain the category of the element.

(b) State the properties of the element that suit its applications in the daily life.

(c) Advise him on the side effect of the material to the environment and how it can be solved.

SECTION B
Attempt one item

Item 2

Air is a mixture of several gases constituting the earth's atmosphere and it is one of the natural resources gifted by God which needs to be conserved.

The residents of Busitema village in Busia district carry out charcoal burning as their main activities to earn a living. However, this human activity has drastically caused a change in the atmospheric air quality. This has forced the National Environmental Management Authority (NEMA) to intervene through organizing a meeting to sensitize the residents about the category and the compositions of the natural resource affected, side effect of the human activity and how it can be overcome and also the benefit of conserving a natural resource.

You have been appointed as the representative of the Environmental Educational Officer to enlighten the community member

Task

As a learner of chemistry, come up with a write-up you will use during the meeting.

Item 3

Water bodies such as lakes and rivers hold a large volume of water as a natural resource that has to be preserved.

Many residents of Bwondha town council in Mayuge district dispose their wastes into the water bodies due to lack of latrines. This has caught the attention of District Health officer (DHO) through the ministry of health together with National Environmental Management Authority (NEMA) to educate the locals about the category and composition of the natural resource, side effect of the human activity, how it can be reduced and why they should preserve it. The meeting has been organized where you have been selected to represent the officials from NEMA.

Task

As a learner chemistry, come-up with short notice you will use

END
CHEM SAMZ